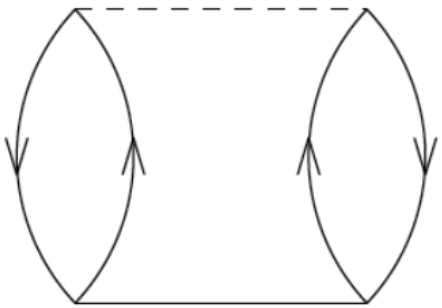
Highly Accurate Molecular Properties using High-Order Relativistic Coupled Cluster Theory

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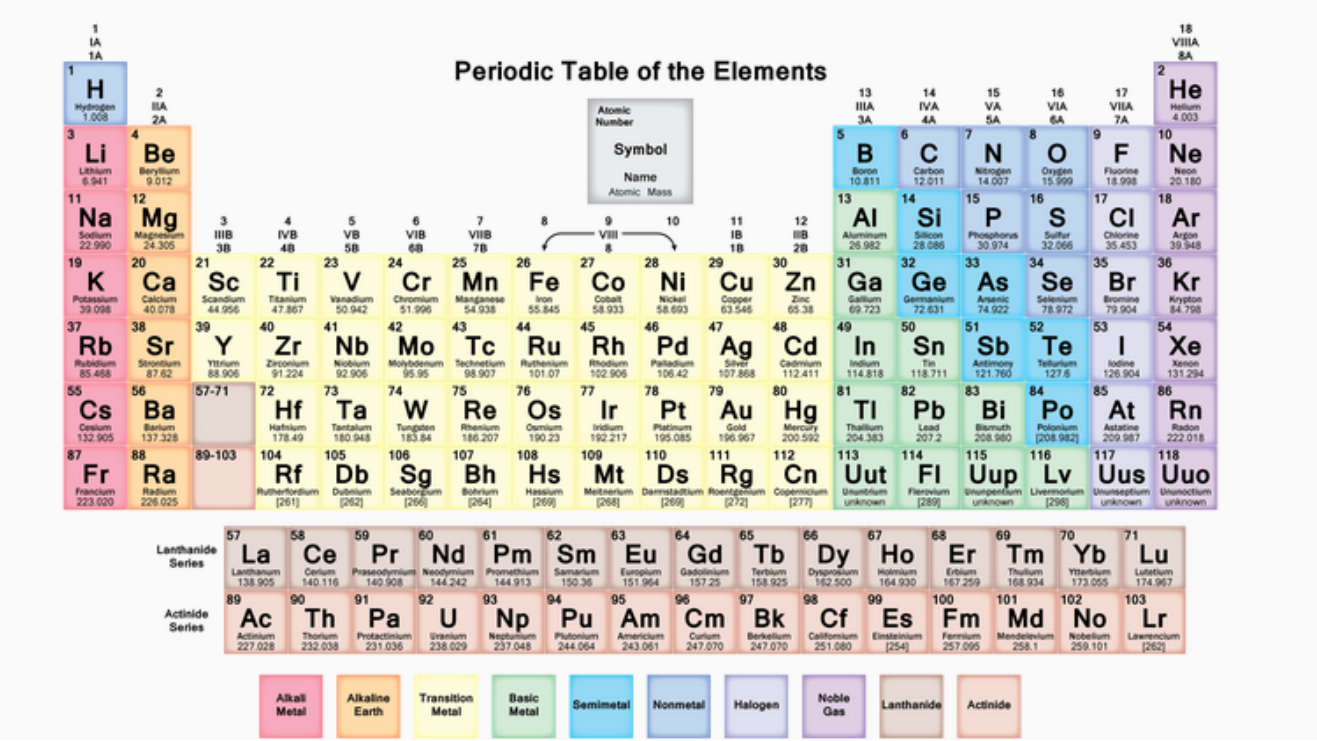
Laboratoire de Chimie et Physique Quantiques - UMR5626

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HAMP-vQED project

Our playground



- Heavy elements need the **theory of special relativity** to be understood.
- HAMP-vQED project will investigate the possible role of **quantum electrodynamics (QED)** in chemistry.
- High accuracy requires **high-order CC methods** and **QED effects**.

Coupled Cluster Theory

- CC ansatz

Excitation operators

Amplitudes

$$|\Psi_{\rm CC}\rangle = e^{\hat{T}}\,|\Phi_0\rangle\,, \quad \hat{T} = \sum_{\mu} t_\mu \quad \hat{\tau}_\mu \tag{1}$$

- Similarity Transformation

$$\hat{\overline{H}}\,|\Phi_0\rangle = E_{\rm CC}\,|\Phi_0\rangle\,, \quad \hat{\overline{H}} = e^{-\hat{T}}\,\hat{H}_N e^{\hat{T}} \tag{2}$$

- Projected equations

$$\langle\Phi_0|: \quad \langle\Phi_0|\hat{\overline{H}}|\Phi_0\rangle = E_{\rm CC} \tag{3}$$

$$\langle\Phi_\mu|: \quad \langle\Phi_\mu|\hat{\overline{H}}|\Phi_0\rangle = 0 \tag{4}$$

- A wide variety of approximations
  - CCD, CCSD, CCSDT, CCSDTQ, ...

Difficulties in High-order CC

1. Cumbersome equations;
2. Designing intermediates with respect to a given cost function;

$$R_i^a = \sum_{jk,bc} A_{bc}^{jk} B_j^b C_{ik}^{ac}, \quad \mathcal{O}(o^3v^3) \tag{5}$$

$$I_c^k = \sum_{jb} A_{bc}^{jk} B_j^b, \quad \mathcal{O}(o^2v^2) \tag{6}$$

$$R_i^a = \sum_{kc} I_c^k C_{ik}^{ac}, \quad \mathcal{O}(o^2v^2) \tag{7}$$

3. Debugging challenges.

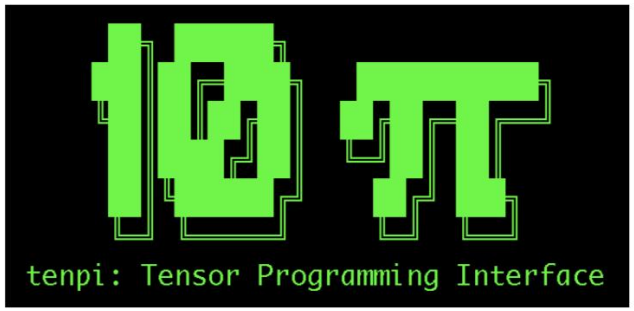
Debugging

verb. [dē'bəg-ing]

1. Being the detective in a crime movie where you are also the murderer.

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J. Brandejs, J. Pototschnig and T. Saue, (2024). Generating coupled cluster code for modern distributed memory tensor software. arXiv:2409.06759.



P rogram
• for
A tomic
• and
M olecular

D irect
I terative
R elativistic
A ll-electron
C alculations



T. Saue et al., *J. Chem. Phys.*, **152**, (2020).

- General-order CC generator.
- Automatic optimization of intermediates.
- Export compatible code (Fortran,...) with calls to tensor libraries (exaTensor, TALSH,..).
- Well integrated within the DIRAC program package.

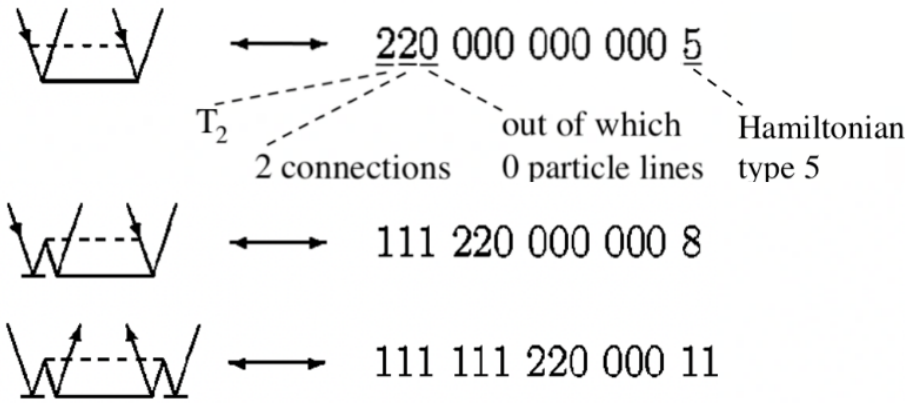
Functionalities

1. General-order CC;
2. Perturbative T,Q;
3. Matrix elements;
4. **Properties?**

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Equation generator principle

- Based on the diagrammatic scheme of M. Kállay and P. R. Surján
  - M. Kállay and P. R. Surján, *J. Chem. Phys.*, **115**, 2945 (2001).
- Integer strings of length 13 are used to represent a diagram.



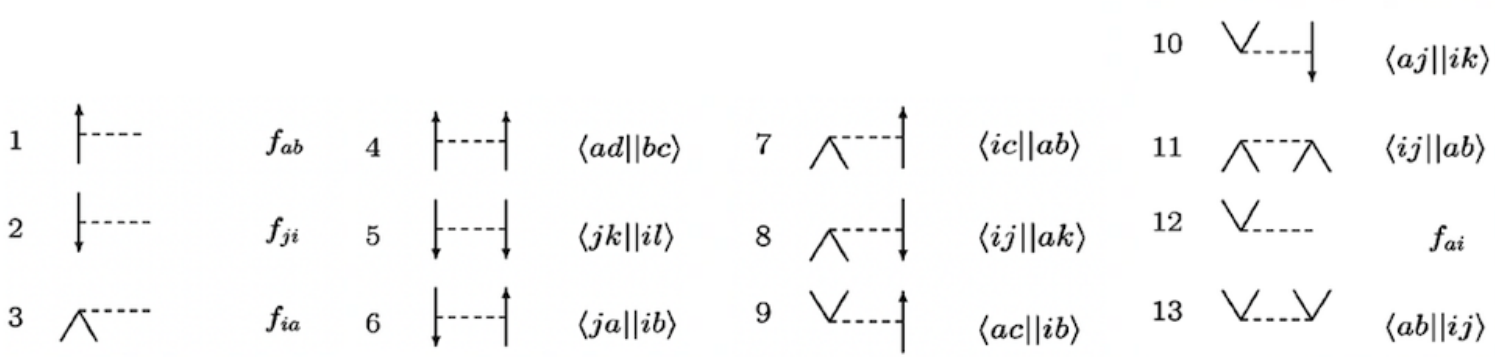
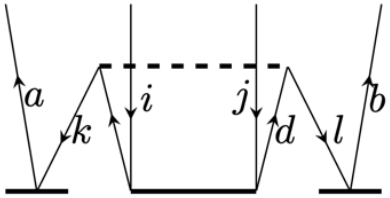


Figure 1: From M. Kállay and P. R. Surján, *J. Chem. Phys.*, **115**, 2945 (2001)

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workflow

Generate only unique diagrams



Translate to line graph representation

```
diag.lines.append([node1, node2])
```

Plot diagrams PDF

Apply rules to obtain equation terms

$$\frac{1}{4}\, P(ab)\, V_{cd}^{kl}\, t_k^a\, t_l^b\, t_{ij}^{cd}$$

Plot PDF

Apply permutations

Devise intermediates (op. count)

$$D_1{}_i^k = f_c^k\, t_i^c$$

$$r_{ji}^{ba} = t_{jk}^{ba}\, D_1{}_i^k$$

'Remove' selected intermediates

Test correctness: brute-force contr.

```
for h1 in range (0, 0):
 for p5 in range (0, V):
  temp_4[h4][h5][h2][h1] +=
  t_vo[p5][h2] * temp_3[h4][h5][p5][h1]
```

Generate code: instructions for tensor lib.

```
talsh_tensor_contract(R2,T2,D1,
"R2(b1,a1,j1,i1)+=T2(b1,a1,j1,k1)*D1(k1,i1)")
```

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Correctness of energies

method	code	convergence	Total energy [E_h]
SCF	DIRAC SCF	1.4E-12[a]	-7.97 9321 565
SCF	MRCC[1]	6.2E-15[a]	-7.97 9321 563
MP2	DIRAC ExaCorr		-7.99 1935 061
MP2	MRCC		-7.99 1935 059
CCSD	DIRAC tenpi[2 b]	0.6E-09[c]	-7.99 8346 943
CCSD	MRCC	2.9E-10[c]	-7.99 8346 941
CCSDT	DIRAC tenpi[b]	0.5E-09[c]	-7.99 8358 347
CCSDT	MRCC	4.2E-10[c]	-7.99 8358 344
CCSDTQ	DIRAC tenpi[b]	0.3E-05[c]	-7.99 8358 347
CCSDTQ	MRCC	7.0E-10[c]	-7.99 8358 363

Table 1: Correctness check on LiH. CAS(4,11). Calculated on Olympe. 6-31G basis. [a]final energy difference, [b]using the ExaTENSOR library, [c]final norm of the residual vector.

[1] M. Kállay et al., MRCC, a quantum chemical program suite, 2024. See `www.mrcc.hu`.

[2] J. Brandejs, J. Pototschnig and T. Saue, (2024). Generating coupled cluster code for modern distributed memory tensor software. arXiv:2409.06759.

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Expectation Values

- In CC theory properties are usually defined using the Lagrangian framework[3]

$$L_{\mathrm{CC}}(\boldsymbol{t},\bar{\boldsymbol{t}},\ \boldsymbol{\varepsilon}\ ) = \langle\Phi_0|(1+\hat{\Lambda})\hat{\overline{H}}|\Phi_0\rangle\,, \quad \hat{\Lambda} = \sum_{\mu} \bar{t}_{\mu}\ \hat{\tau}^{\dagger}_{\mu} \tag{8}$$

Field strength; Multipliers; De-excitators

- Differentiating it

$$\left.\frac{dL_{\mathrm{CC}}(\boldsymbol{t},\bar{\boldsymbol{t}},\boldsymbol{\varepsilon})}{d\varepsilon_A}\right|_{\boldsymbol{\varepsilon}=\mathbf{0}} = \left[\frac{\partial L_{\mathrm{CC}}(\boldsymbol{t},\bar{\boldsymbol{t}},\boldsymbol{\varepsilon})}{\partial \varepsilon_A} + \frac{\partial L_{\mathrm{CC}}(\boldsymbol{t},\bar{\boldsymbol{t}},\boldsymbol{\varepsilon})}{\partial t_\nu}\frac{dt_\nu}{d\varepsilon_A} + \frac{\partial L_{\mathrm{CC}}(\boldsymbol{t},\bar{\boldsymbol{t}},\boldsymbol{\varepsilon})}{\partial \bar{t}_\nu}\frac{d\bar{t}_\nu}{d\varepsilon_A}\right]_{\boldsymbol{\varepsilon}=\mathbf{0}} \tag{9}$$

  - We impose the stationarity of the Lagrangian with respect to t_ν and $\bar{t}_\nu$:

$$\frac{\partial L_{\mathrm{CC}}(\boldsymbol{t},\bar{\boldsymbol{t}},\boldsymbol{\varepsilon})}{\partial \bar{t}_\nu} = 0 = \langle\Phi_\nu|\hat{\overline{H}}|\Phi_0\rangle \leftarrow \text{Amplitudes} \tag{10}$$

$$\frac{\partial L_{\mathrm{CC}}(\boldsymbol{t},\bar{\boldsymbol{t}},\boldsymbol{\varepsilon})}{\partial t_\nu} = 0 = \langle\Phi_0|(1+\hat{\Lambda})[\hat{\overline{H}},\hat{\tau}_\nu]|\Phi_0\rangle \leftarrow \text{Multipliers} \tag{11}$$

- First-order properties

$$\left.\frac{dL_{\mathrm{CC}}}{d\varepsilon_A}\right|_{\boldsymbol{\varepsilon}=\mathbf{0}} = \left[\langle\Phi_0|(1+\hat{\Lambda})\frac{\partial\hat{\overline{H}}}{\partial\varepsilon_A}|\Phi_0\rangle\right]_{\boldsymbol{\varepsilon}=\mathbf{0}} = \sum_{pq} h_{pq;A}\ \gamma^{\mathrm{CC}}_{pq} \tag{12}$$

One-body response density matrix; Perturbation

[3] T. Helgaker and P. Jørgensen, *Theor. Chim. Acta*, **75**, 111-127 (1989).

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Correctness of dipole moments

method	**code**	**convergence**	**Total dipole moment** $[\mathrm{E}_h a_0^{-1}]$
SCF	DIRAC SCF		-2.33629085
SCF	MRCC[4]		-2.33629086
CCSD	DIRAC tenpi[a]	0.5×10^{-9}	-2.16794694
CCSD	DIRAC[b]	0.5×10^{-9}	-2.16794694
CCSD	MRCC	1.3×10^{-10}	-2.16794696
CCSDT	DIRAC tenpi	0.4×10^{-9}	-2.16768731
CCSDT	MRCC	4.9×10^{-10}	-2.16768734

Table 2: Correctness check on LiH. CAS(4,11). Calculated on Olympe. 6-31G basis. [a] using the TALSH library, [b] handwritten code.

[4] M. Kállay et al., MRCC, a quantum chemical program suite, 2024. See `www.mrcc.hu`.

Conclusions and Perspectives

Conclusions

1. CCSDT and CCSDTQ are now available in DIRAC.
2. CCSDT expectation value is now available in DIRAC.

Perspectives

1. CC development
   - CCSDTQ, CCSD(T) and CCSDT(Q) expectation values.
   - Orbital relaxation.
2. Molecular Properties
   - **Electric Field Gradient** $\overset{\text{NQCCs}}{\Rightarrow}$ Nuclear Quadrupole Moments.
   - Magnetic properties require the use of **negative energy solutions** of the Dirac equation.
   - Core molecular properties: **QED-effects** may play an important role.

HAMP-vQED

Highly Accurate Molecular Properties using Variational Quantum Electrodynamics







HAMP-vQED team











From the left: Jan Brandejs, Gabriele Fabbro, Trond Saue, Ryan Benazzouk and Dávid Ferenc.